

ITIS 4250
Efren Antonio
Lab #3
October 19, 2024

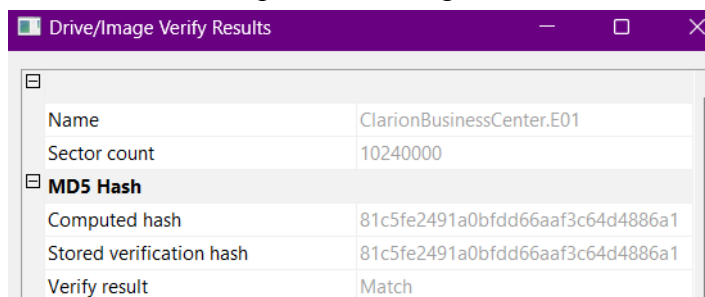
Overview

On October 19, 2024, Det. Lansbury of the UNCC Police Department provided me with two forensic images for analysis: one from a laptop belonging to Malcom Monarch and another from a computer at the Clarion Hotel Business Center. Monarch submitted his laptop for repair at Conjectural Technologies where technician Pete White reported the device that was later obtained by law enforcement with a warrant. The image of the hotel lobby computer was obtained from the Clarion Hotel Internet kiosk on consent. The goal of this examination is to assist my investigator Agent Sampson gather evidence related to Monarch's activities, including communications, web searches, media, and other artifacts that may be linked to criminal activity.

Exam Preparation

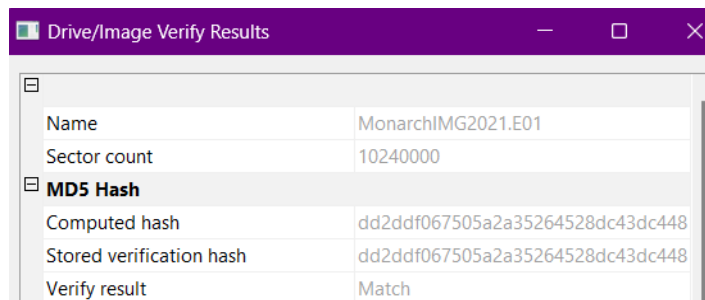
The examination environment used is a HP Laptop running Microsoft Windows 11 Home Version 10.0.22631. The tool used for verification of the image is Forensic Tool Kit (FTK) Imager Version 4.7.1.2. The tool used for analyzing communications and web searches is Autopsy Version 4.21.0. The tool to find the OS version was Registry Explorer Version 1.1.0.6

The verified hash of ClarionBusinessCenter.E01 is **81c5fe2491a0bfdd66aaf3c64d4886a1** provided in Figure 1 below. The verified hash of MonarchIMG2021.E01 is **dd2ddf067505a2a35264528dc43dc448** provided in Figure 2 below.



Drive/Image Verify Results	
Name	ClarionBusinessCenter.E01
Sector count	10240000
MD5 Hash	
Computed hash	81c5fe2491a0bfdd66aaf3c64d4886a1
Stored verification hash	81c5fe2491a0bfdd66aaf3c64d4886a1
Verify result	Match

Figure 1: ClarionBusinessCenter.E01 FTK Imager MD5 Hash Verification



Drive/Image Verify Results	
Name	MonarchIMG2021.E01
Sector count	10240000
MD5 Hash	
Computed hash	dd2ddf067505a2a35264528dc43dc448
Stored verification hash	dd2ddf067505a2a35264528dc43dc448
Verify result	Match

Figure 2: MonarchIMG2021.E01 FTK Imager MD5 Hash Verification

Analysis

1. Was there communication from someone claiming to be the Guild's operative, or can you identify the operative's name/email?

There was email communication from someone claiming to be the Guild's operative. This person was identified as **Hamilton J. Fantomos aka "The Phantom Limb" with the email: generalconsolidatedindustries@aol.com**. The email can be seen in Figure 3 below.

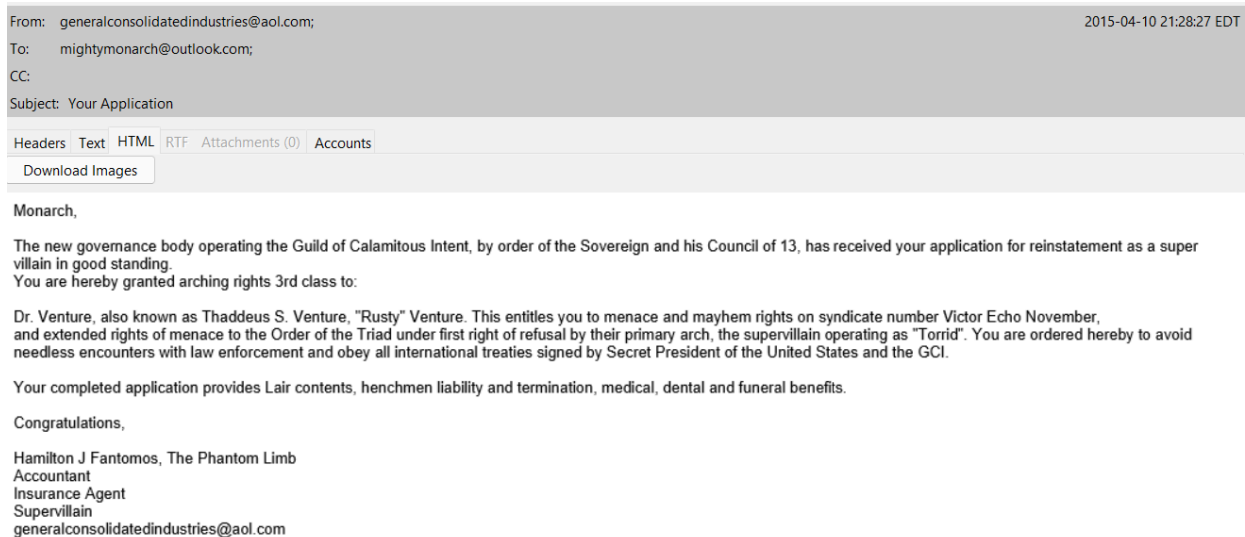


Figure 3: Email From The Guild Operative

2. What is Malcolm's email address? Did he send email to someone other than the guild operative? Who?

Malcolm's email address is: **mightymonarch@outlook.com**. He sent emails to someone else named **Sheila with the email: doctormrsthemonarch@outlook.com**. This can be seen in Figure 4 below.

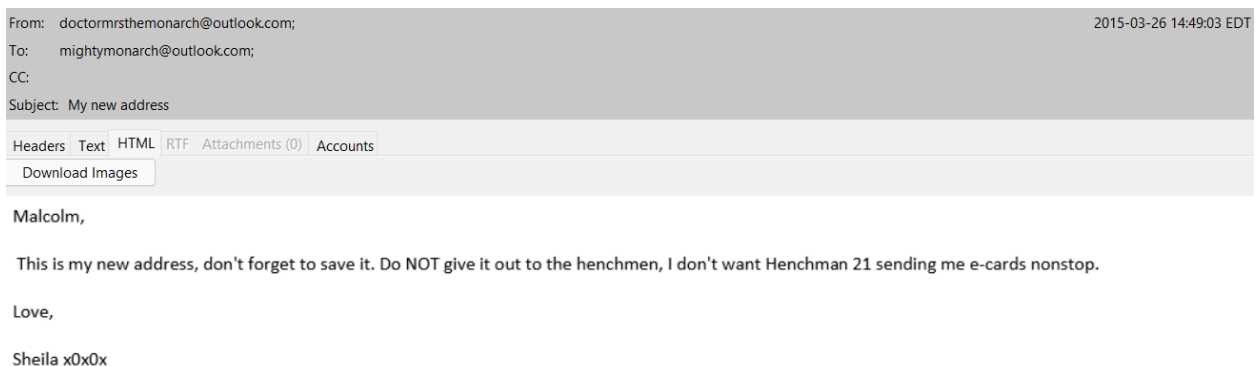


Figure 4: Email From Sheila

3. What version OS was running on the computer in use?

The computer in use was running **Windows 7 Enterprise Version 6.1**. This was found using Registry Explorer on the SOFTWARE file in the RegistryM folder seen in Figure 5 below.

Value Name	Value Type	Data	Value Slack
CurrentBuildNumber	RegSz	7601	74-00
CurrentType	RegSz	Multiprocessor Free	65-00-64-00-00-00-00-...
CurrentVersion	RegSz	6.1	33-00-32-00
DigitalProductId	RegBinary	A4-00-00-00-03-00-00-0...	
DigitalProductId4	RegBinary	F8-04-00-00-04-00-00-0...	00-00-00-00
EditionID	RegSz	Enterprise	30-01-10-63-30-01
InstallationType	RegSz	Client	00-00-00-00-00-00
InstallDate	RegDword	1382545364	
PathName	RegSz	C:\Windows	65-72-00-00-00-00
ProductId	RegSz	00392-972-8000024-85319	00-00-00-00
ProductName	RegSz	Windows 7 Enterprise	30-01
RegisteredOrganization	RegSz		

Figure 5: Registry Explorer Value Table For Current Windows Version

4. Did Malcolm ever search for any terms that might mean he was going to cause any explosions?

Malcolm searched for a few terms that might mean he was going to cause explosions. These search terms included **“how to make fireworks”** and **“is it illegal to make a bomb”**. He also bookmarked a **bomb-making tutorial website**. These are all provided in Figures 6, 7, 8 below.

Web Search		is it illegal to learn to make a bomb and such stuff??reposted 10 pts.? Yahoo Answers 1SPS Visited: IEUser@https://answers.yahoo.com/question/index?qid=20110113225304AAWIXLD 1SPS is it illegal to learn to make a bomb and such stuff??reposted 10 pts.? Yahoo Answers 1SPS :2015032920150330: IEUser@https://answers.yahoo.com/question/index?qid=20110113225304AAWIXLD 2015032920150330: IEUser@:Host: answers.yahoo.com 1SPS is it illegal to learn to make a bomb and such stuff??reposted 10 pts.? Yahoo Answers 1SPS
Source		
Term:	how to make fireworks	
Time:	2015-03-30 00:22:41 EDT	
Domain:	bing.com	
Program Name:	Microsoft Edge Analyzer	
Host:	MonarchIMG2021.E01_6992 Host	
Data Source:	MonarchIMG2021.E01	

Figure 6 & 7: Search Results Related To Creating Explosions

Bookmark Details	
Title:	Bomb-making instructions on the internet - Wikipedia, the free encyclopedia.url
Date Created:	2015-03-29 20:19:08 EDT
Domain:	wikipedia.org
URL:	http://en.wikipedia.org/wiki/Bomb-making_instructions_on_the_internet
Program Name:	Internet Explorer Analyzer
Source	
Host:	MonarchIMG2021.E01_6992 Host
Data Source:	MonarchIMG2021.E01

Figure 8: Bookmarked Bomb Making Instructions

5. How about how to cover his tracks?

In order to cover his tracks, Malcolm also searched “**how to hide evidence**”, “**how to hide evidence computer**”, and “**ccleaner**” which is software that helps cover his tracks. These searches can be seen in Figure 9 below.

bing.com	how to hide evidence	Microsoft Edge Analyzer	2015-03-29 23:19:12 EDT	MonarchIMG2021.E01
bing.com	how to hide evidence computer	Microsoft Edge Analyzer	2015-03-29 23:19:16 EDT	MonarchIMG2021.E01
bing.com	ccleaner	Microsoft Edge Analyzer	2015-03-29 23:19:47 EDT	MonarchIMG2021.E01

Figure 9: Search Terms to Help Cover His Tracks

6. Explain in detail what he was doing, with what software and when.

Malcolm was using CCleaner, an anti-forensic software that is used to cover his tracks and make it difficult to extract more evidence. CCleaner was used to wipe free space drive C:\. The last time the CCleaner was modified was 3-30-2015 at 7:50 GMT.

Hex	Text	Application	File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences
CCleaner									
Metadata									
Name: CCleaner									
Number of subkeys: 0									
Number of values: 17									
Modification Time: 2015-03-30 00:07:50 GMT+00:00									
Values									
Name	Type	Value							
Language	REG_S...	1033							
SkipUAC	REG_S...	1							
UpdateKey	REG_S...	03/29/2015 05:07:37 PM							
WipeFreeSpaceDrives	REG_S...	C:\							
CookiesToSave	REG_S...	*.piriform.com facebook.com google.com yahoo.co...							
RunICS	REG_S...	0							
Monitoring	REG_S...	1							
SystemMonitoring	REG_S...	1							

Figure 10: Registry Explorer Value Table For CCleaner

7. Are there any photos that might have been downloaded? How do you know the photo was downloaded and not just moved to the folder where it was found? Is there a photo at this location that might tell us how he appears?

The photos that have been downloaded are “images.jpg”, “venture-bros-monarch.jpg”, and “the_monarch_wants_you.jpg.” These photos were downloaded and not moved into the folder because they appear in the web downloads table of Autopsy with the download path and download link. In these photos, we can see he appears to have long red facial hair, a gold crown, and a yellow suit on. These can be seen in Figures 11 and 12 below.

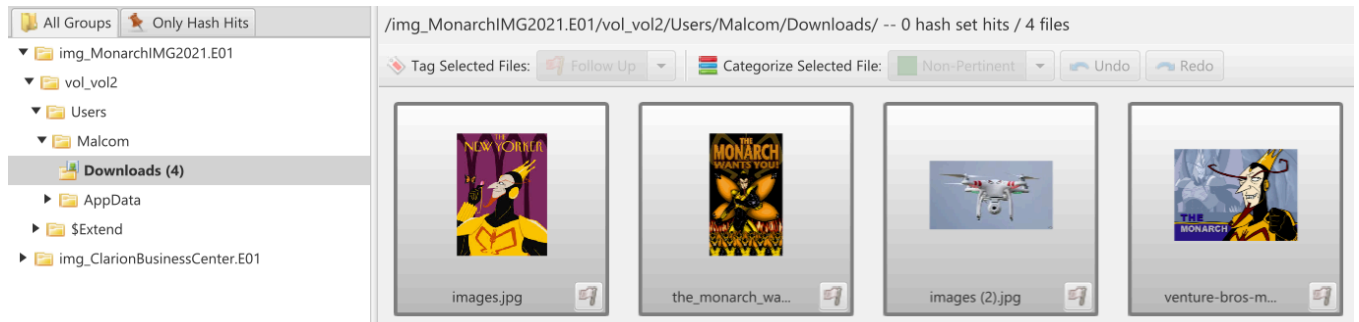


Figure 11: Photos in the Downloads Folder

Web Downloads						
Table Thumbnail Summary						
Source Name	S	C	O	Path	URL	
A green crown m af9f6fad-b6a4-4f31-ad6f-ad649f				/Downloads/NonImportantStuff/A green crown m af9f6	aboutinternet	
History			2		https://www.google.com/url?sa=i&rct=j&q=&esrc=s&...	
History			2	C:\Users\IEUser\Downloads\the_monarch_wants_you_...	http://files.chesscomfiles.com/images_users/tiny_mce/...	
History			2		https://www.google.com/url?sa=i&rct=j&q=&esrc=s&...	
History			2	C:\Users\IEUser\Downloads\images.jpg	https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd...	
History			2	C:\Users\IEUser\Downloads\venture-bros-monarch.jpg	http://www.venturebrosblog.com/wp-content/uploads/	

Figure 12: Web Downloads Table In Autopsy

8. Did Malcolm do any research into forensics software? What specifically?

Malcolm looked into **FTK Imager**, specifically he researched how to uncover hidden evidence using this software. He also looked into a whitepaper explaining how to uncover hidden evidence using other computer forensics tools. These can be found in Figures 13 and 14 below.

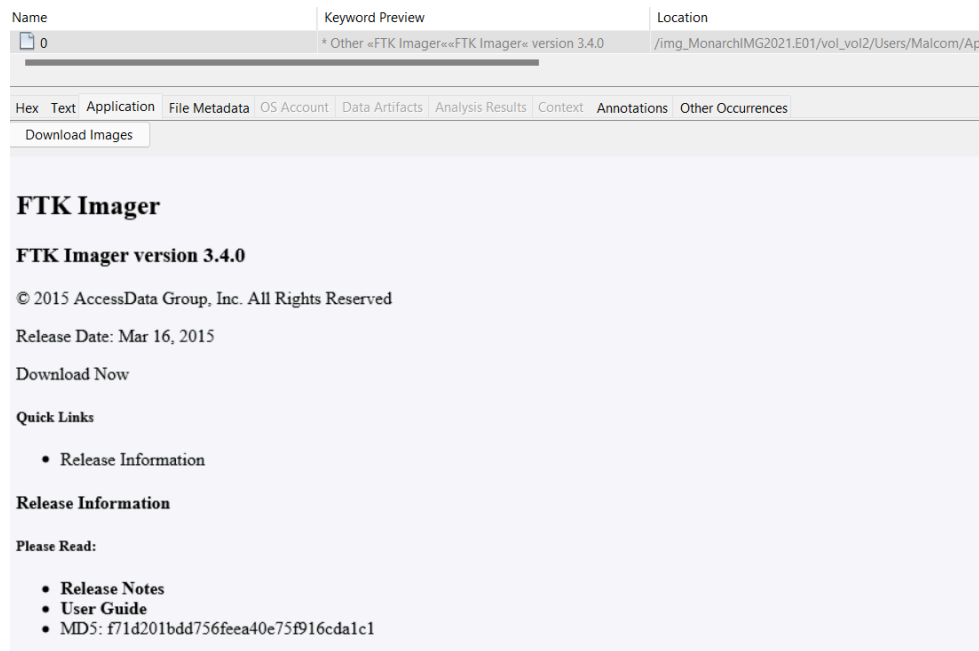


Figure 13: FTK Imager Version 3.4.0 Homepage and Download Link

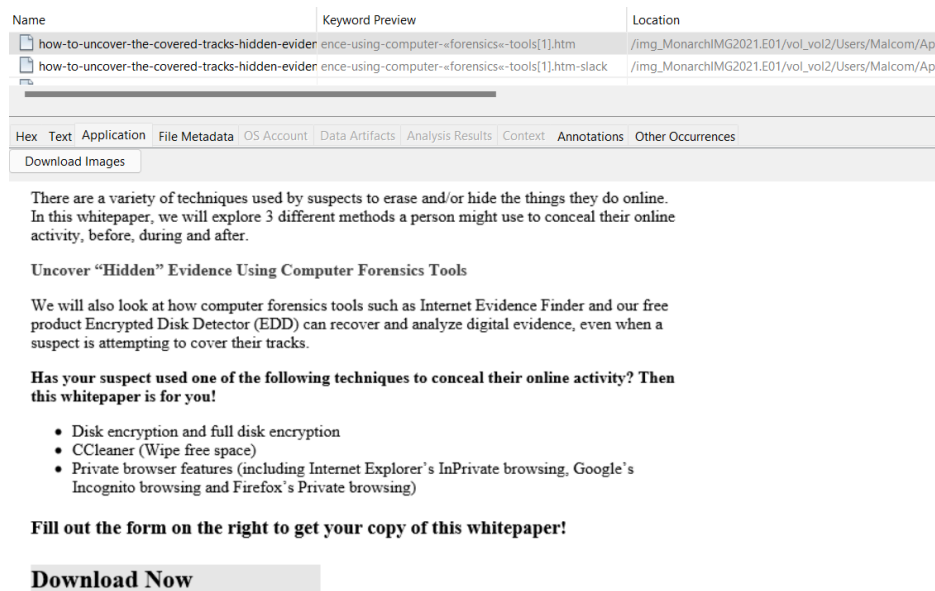


Figure 14: Whitepaper Malcolm Researched

9. Is there any evidence he may have opened photos locally with an internet browser? Are those photos still available on the system?

In the web browsing history in Autopsy, there is evidence that Malcolm Monarch opened three images locally using an internet browser. **These entries are highlighted in Figure 15 below and have the file paths reference local directories instead of URLs pointing to web servers. These photos are still available on the system and can be seen in Figure 16 below.**

Web History						
Table Thumbnail Summary						
Source Name	S	C	O	URL	Date Accessed	Program Name
WebCacheV01.dat				about:blank	2015-04-11 01:46:32 EDT	Microsoft Edge Analyzer
WebCacheV01.dat				about:blank	2015-03-30 00:17:20 EDT	Microsoft Edge Analyzer
WebCacheV01.dat				file:///C:/Users/IEUser/Downloads/images.jpg	2015-03-30 00:03:48 EDT	Microsoft Edge Analyzer
WebCacheV01.dat				file:///C:/Users/IEUser/Downloads/the_monarch_wants_...	2015-03-30 00:06:22 EDT	Microsoft Edge Analyzer
WebCacheV01.dat				file:///C:/Users/IEUser/Downloads/venture-bros-monar...	2015-03-29 23:59:31 EDT	Microsoft Edge Analyzer

Figure 15: Autopsy Web History Highlighting Three Locally Opened Photos

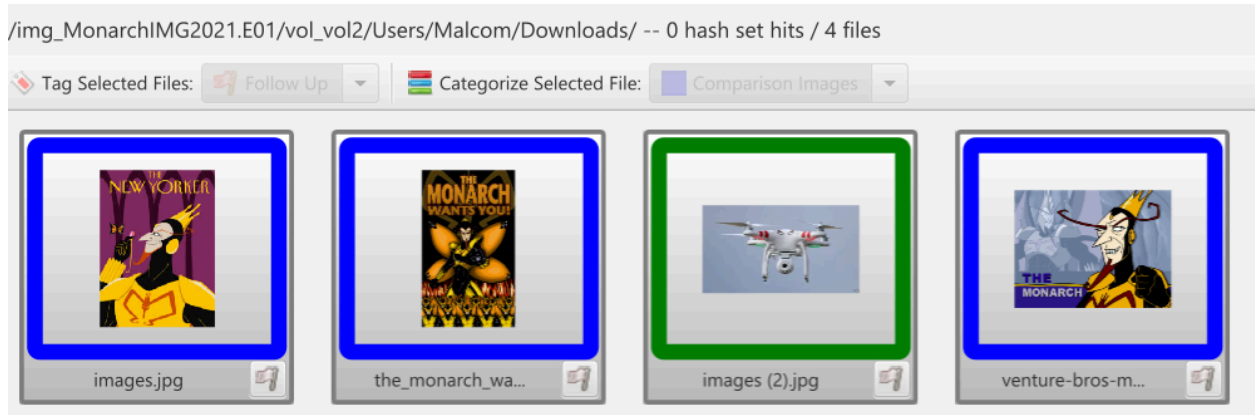


Figure 16: Locally Opened Photos Highlighted in Blue

Conclusion

The forensic analysis of Malcom Monarch's laptop and the Clarion Hotel Business Center computer provided key insights into his activities. It revealed communications with a Guild operative along with searches related to fireworks, bombs, and using tools like CCleaner to remove traces of activity. Monarch also explored forensic software, potentially to understand how evidence might be uncovered. Downloaded images and web history offer details about his appearance and browsing behavior. These findings provide valuable information for further examination of Monarch's actions and intent.

Name	ClarionBusinessCenter.E01
Sector count	10240000
MD5 Hash	
Computed hash	81c5fe2491a0bfdd66aaf3c64d4886a1
Stored verification hash	81c5fe2491a0bfdd66aaf3c64d4886a1
Verify result	Match

Name	MonarchIMG2021.E01
Sector count	10240000
MD5 Hash	
Computed hash	dd2ddf067505a2a35264528dc43dc448
Stored verification hash	dd2ddf067505a2a35264528dc43dc448
Verify result	Match

Figure 17: Reverified MD5 Hash Values After The Examination

Signed:

Efren Antonio